Towards Fast and High-Quality Sign Language Production

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ABSTRACT

Sign Language Production (SLP) aims to automatically translate a spoken language description to its corresponding sign language video. The core procedure of SLP is to transform sign gloss intermediaries into sign pose sequences (G2P). Most existing methods for G2P are based on sequential autoregression or sequence-to-sequence encoder-decoder learning. However, by generating target pose frames conditioned on the previously generated ones, these models are prone to bringing issues such as error accumulation and high inference latency. In this paper, we argue that such issues are mainly caused by adopting autoregressive manner. Hence, we propose a novel Non-AuToregressive (NAT) model with a parallel decoding scheme, as well as an External Aligner for sequence alignment learning. Specifically, we extract alignments from the external aligner by monotonic alignment search for gloss duration prediction, which is used by a length regulator to expand the source gloss sequence to match the length of the target sign pose sequence for parallel sign pose generation. Furthermore, we devise a spatial-temporal graph convolutional pose generator in the NAT model to generate smoother and more natural sign pose sequences. Extensive experiments conducted on PHOENIX14T dataset show that our proposed model outperforms state-of-the-art autoregressive models in terms of speed and quality.

CCS CONCEPTS

• **Computing methodologies** → **Motion capture**; **Activity recognition and understanding**.

KEYWORDS

Sign Language Production; Non-Autoregressive Generation; Deep Learning

ACM Reference Format:
Wencan Huang, Wenwen Pan, Zhou Zhao, and Qi Tian. 2021. Towards Fast and High-Quality Sign Language Production. In *Proceedings of the 29th ACM International Conference on Multimedia (MM '21), October 20–24, 2021, Virtual Event, China.* ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3474085.3475463

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MM '21, October 20–24, 2021, Virtual Event, China
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ACM ISBN 978-1-4503-8651-7/21/10…$15.00
https://doi.org/10.1145/3474085.3475463

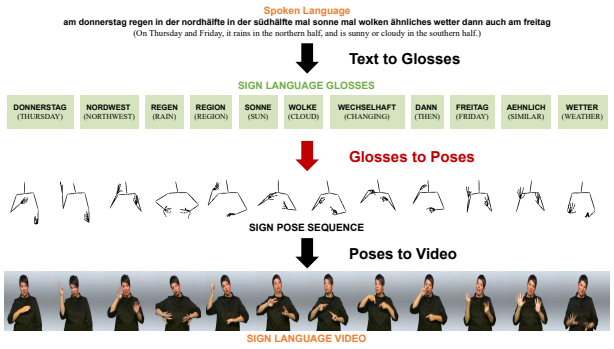


Figure 1: The typical pipeline of sign language production, in which we mainly focus on the G2P problem.

1 INTRODUCTION

Sign language, a rich visual language with complex grammatical structures, is the language of communication for the Deaf community. To involve the Deaf in the predominantly spoken language of the wider world, a large amount of methods [38–40, 43] have been recently proposed to tackle the challenging Sign Language Production (SLP) problem. Given a spoken language description, SLP aims to automatically translate it into the corresponding continuous sign sequence. Generally, sign sequences can be represented as sign skeleton pose sequences [38, 40] or sign language videos [39].

As shown in Figure 1, previous works for SLP usually translate the spoken language into gloss[1] intermediary first, then generate sign pose sequence from the gloss sequence (G2P) [38, 40], and finally maybe use the sign pose sequence to drive a style image to generate a sign language video [39]. Thereinto, G2P, the core procedure of this pipeline, is the focus of this paper.

In general, existing approaches for G2P are based on sequential autoregression or sequence-to-sequence encoder-decoder learning, which generates the next pose frame depending on the gloss sequence as well as the current generated pose frame [38, 40]. However, these autoregressive methods are likely to face the following two challenges: 1) Error accumulation: the prediction accuracy of

[1]Sign glosses are spoken language words that match the meaning of signs and, linguistically, manifest as minimal lexical items.

the current frame relies heavily on that of previous frames, resulting in the recurrent prediction error propagation over time [40]. 2) High inference latency: the processing time increases approximately linearly with the length of the sign pose sequence. Prior work PT [40] alleviates the error accumulation problem with various data augmentation approaches, but it still has an unsatisfying performance in terms of speed and quality due to autoregressive generation.

To solve these problems, non-autoregressive models [28, 49] have been proposed in the skeleton sequence generation field. The most typical one is mNAT [28] for human motion prediction. mNAT eases the aforementioned issue and improves performance by predicting all future skeleton frames in parallel via a non-autoregressive frame decoder, which takes a positional encoding and a context feature from a context encoder as inputs. However, different from human motion prediction problem, G2P also requires modeling of alignments between sign pose sequences and gloss sequences.

To this end, we propose a novel Non-AuToregressive (NAT) model with a complete non-autoregressive decoding scheme, as well as an External Aligner (EA) for sequence alignment learning. We call our whole system NAT-EA. Specifically, the NAT is composed of feed-forward transformers, a length regulator and a spatial-temporal graph convolutional pose generator. The length regulator contains a duration predictor. In inference stage, gloss embedding sequences are first transformed into gloss-synchronized feature sequences by a feed-forward transformer. Next, the duration predictor predicts the lengths of the sign pose sequences that are aligned with each gloss. According to the lengths, gloss-synchronized feature sequences are expanded to pose-synchronized feature sequences by the length regulator, which are then transformed to the target sign pose sequences by another feed-forward transformer and the spatial-temporal graph convolutional pose generator successively. In training stage, the duration predictor is optimized towards the target lengths, which are generated by a well-trained EA.

Next, we consider the concrete design of the EA. In the field of non-autoregressive text-to-speech synthesis, targets of duration predictors generally come from well-trained autoregressive models [37]. However, the autoregressive models for G2P problem can hardly get satisfactory alignments between sign pose sequences and gloss sequences due to the error accumulation problem. Therefore, inspired by AlignTTS [55], we apply a mix density network [5] to transform gloss embedding sequences into gloss-synchronized sequences of sign pose Gaussian distributions, which are then aligned with target sign pose sequences by monotonic alignment search [25] to get the alignment lengths of each gloss, which are used as the target lengths of the duration predictor in training stage.

Spatial-temporal graph convolution is widely-used in skeleton sequence generation [49], which makes the generated skeleton sequences smoother and more natural to improve the model performance. Hence, we devise a spatial-temporal graph convolutional pose generator for our NAT model, consisting of multiple spatial-temporal graph convolution layers and graph up-sampling layers, which transforms pose-synchronized feature sequences into target sign pose sequences.

Our main contributions can be summarized as follows:

- We propose a novel Non-AuToregressive model with External Aligner (NAT-EA) for the G2P problem in sign language production, which improves the model performance by resolving the error accumulation problem and the sequence alignment problem, and speeds up the procedure by generating sign pose sequences parallelly. This is the first non-autoregressive model for the G2P problem.
- We devise the External Aligner to generate the alignment lengths of each gloss, which are used as the target lengths of the duration predictor in training stage, and develop the spatial-temporal graph convolutional pose generator to generate target sign pose sequences.
- Extensive experiments on the challenging PHOENIX14T dataset [7] show the effectiveness of our proposed NAT-EA method.

2 RELATED WORK

2.1 Human Motion Modeling

It is challenging to model human motion dynamics due to the stochastic nature and spatio-temporal complexity. A large family of the existing work [10, 16, 51, 53] formulates motion dynamics as a sequence of 2D or 3D body keypoints, thanks to the success of human pose estimation [9]. Most of these approaches use autoregressive models to generate a motion sequence from a static image or previous motions, and the accumulation of errors is usually diminished by denoising strategies [16]. Some other methods consider generating the entire sequence directly in a non-autoregressive way [28, 49] for human motion prediction. In this work, we introduce a non-autoregressive model to sign language production.

2.2 Sign Language Production

Sign language has been a focus of computer vision researchers for over 30 years [2, 41, 45], primarily on isolated Sign Language Recognition (SLR) [32], Continuous Sign Language Recognition (CSLR) [6, 26] and the more demanding task of Sign Language Production (SLP). Previous approaches to SLP have extensively used animated avatars [23, 31] that can generate realistic sign production, but rely on phrase lookup and pre-generated sequences. Statistical Machine Translation (SMT) has also been applied to SLP [24, 27], relying on static rule-based processing that can be difficult to encode. Recently, deep learning approaches have been applied to the task of SLP [13, 42, 48]. Stoll et al. [43] propose the first deep SLP system, which adopts the typical three-step SLP pipeline. In the G2P procedure, they produce a concatenation of isolated 2D skeleton poses [15] mapped from sign glosses via a look-up table. Following this pipeline, B. Saunders et al. [40] propose the progressive transformer to learn the mapping between glosses and skeleton pose sequences directly. Further, they proposed SIGNGAN [39] to produce photo-realistic continuous sign language videos from spoken language, in which they apply a mixture density network to boost the performance of their G2P model.

However, all these methods for G2P problem are based on autoregressive models, which suffers from the error accumulation problem and high inference latency. Our framework, instead, generates the entire sign pose sequence directly in a non-autoregressive way and resolves these problems effectively.

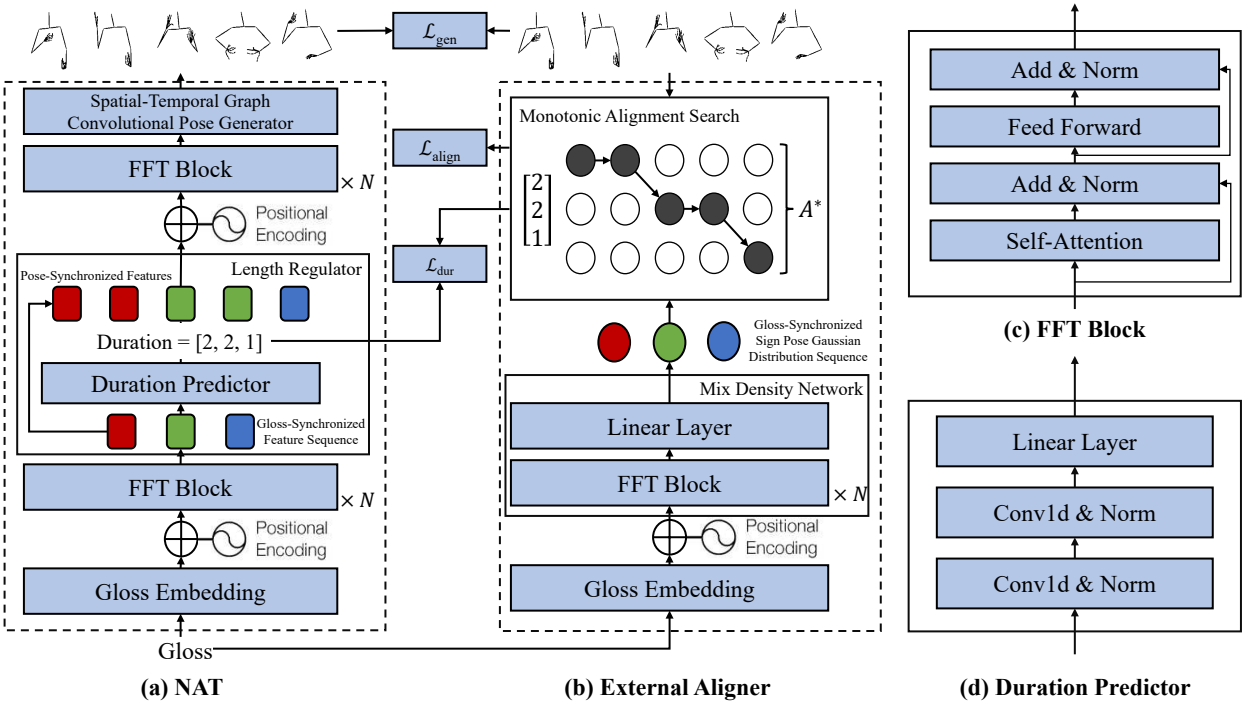


Figure 2: The overall architecture for NAT-EA. (a). The Non-AuToregressive model. (b). The External Aligner. (c). The feed-forward Transformer block. (d). The duration predictor. The alignment loss $\mathcal{L}_{\text{align}}$ only exists in the training process of the EA.

2.3 Non-Autoregressive Models

RNN-based models [12, 22] achieved great success in sequence modeling, especially in Neural Machine Translation (NMT) [1, 11, 44]. In general, these methods generate tokens in an autoregressive way, i.e., the new output word is dependent on the previously generated output. Such left-to-right decoding manner suffers from problems like low efficiency and error accumulation [3, 19]. To remedy these issues, several efforts have been paid on avoiding recurrence in sequence modeling. Gehring et al. [17] proposed a sequence model based entirely on convolutional neural networks. Gu et al. [21] proposed Non-AuToregressive (NAT) transformer that makes use of fertilities which represents how many times each source tokens are copied. Besides the study of NAT NMT [29, 36], recently, many works introduced the NAT decoding into other sequence-to-sequence tasks, such as video caption [52], lipreading [30], speech recognition [46] and speech synthesis [35, 37].

As a sequence modeling task, the G2P problem of the typical SLP pipeline shares similar nature with NMT and meanwhile possesses unique characteristics. 1) G2P is usually formulated as a continuous generation task rather than a discrete one. Hence, several useful tricks like Beam Search [20, 44] in NLP could not be directly adopted. 2) Sign pose sequence contains rich skeletal structure information, which natural language token rarely owns. Therefore, in this paper, we propose a novel NAT-EA model with a spatial-temporal graph convolutional pose generator to achieve non-autoregressive decoding and learn the rich skeletal structure in the meantime.

3 THE PROPOSED METHOD

Given a sign language gloss sequence $\mathbf{X}$, the G2P procedure of the typical SLP pipeline aims to generate the semantically corresponding sign pose sequence $\mathbf{Y} = (\mathbf{y}_1, \mathbf{y}_2, \cdots, \mathbf{y}_{N_Y}) \in \mathbb{R}^{N_Y \times J \times K}$, where each sign pose frame $\mathbf{y}_n = \{\mathbf{y}_n^j\}_{j=1}^{J}$ represents the single skeleton, containing J joints data. $\mathbf{y}_n^j \in \mathbb{R}^K$ is a minimal per-joint representation at the n-th frame and the j-th joint, of which K is the feature dimension which represents human joint data.

3.1 The Overall Architecture Design

We first introduce the overall architecture of our Non-AuToregressive model with External Aligner (NAT-EA). As shown in Figure 2, NAT-EA consists of two networks: a Non-AuToregressive model (NAT) and an External Aligner (EA). We devise the EA to generate the alignment lengths of each gloss, which are used as the target lengths of the duration predictor in training stage, and develop the spatial-temporal graph convolutional pose generator to generate target sign pose sequences.

Concretely, we first transform the input gloss sequence into the gloss-synchronized feature sequence by a Feed-Forward Transformer (FFT) block. The FFT block is composed of stacked self-attention and feed-forward layers, which are the same as those in Transformer [47] and autoregressive SLP model [40]. After feature extraction, we apply a duration predictor to predict the logarithmic duration of each gloss $\mathbf{l}^{\text{pred}} = \{l_i^{\text{pred}}\}_{i=1}^{N_X}$, where N_X is the length

of the input gloss sequence, and l_i^{pred} is the logarithmic length of the sign pose sequence that is aligned with the i-th gloss. The duration predictor consists of a 2-layer 1D-convolutional network with ReLU activation, each followed by the layer normalization and the dropout layer, and an extra linear layer to project the hidden states into the output sequence.

Additionally, in training stage of the NAT model, we also feed the input gloss sequence to a well-trained EA to get the target logarithmic duration of each gloss $\mathbf{l}^{\text{trg}} = \{l_i^{\text{trg}}\}_{i=1}^{N_X}$, where $\sum e^{l_i^{\text{trg}}}{}_{i=1}^{N_X} = N_Y$ and N_Y is the length of the target sign pose sequence. Then we expand the gloss-synchronized feature sequence to the pose-synchronized feature sequence according to $\mathbf{l}^{\text{trg}}$ in training stage or according to $\mathbf{l}^{\text{pred}}$ in inference stage. The expansion is conducted by the length regulator, which can be illustrated by an example as follows. Given a feature sequence $H_{\text{gloss}} = [h_1, h_2, h_3, h_4]$ and the corresponding duration sequence $e^{\mathbf{l}} = [1, 2, 3, 2]$, the expanded feature sequence becomes $H_{\text{pose}} = [h_1, h_2, h_2, h_3, h_3, h_3, h_4, h_4]$. When we get $\mathbf{l}^{\text{pred}}$ and $\mathbf{l}^{\text{trg}}$, the duration loss can be calculated by the mean square error (MSE):

$$\mathcal{L}_{\text{dur}} = \text{MSE}(\mathbf{l}^{\text{pred}}, \mathbf{l}^{\text{trg}}) \tag{1}$$

After feature expansion, the pose-synchronized feature sequence is transformed into the sign pose sequence $\mathbf{Y}_{\text{gen}} \in \mathbb{R}^{T \times J \times K}$ by another FFT block and the spatial-temporal graph convolutional pose generator successively. The generation loss is defined as the mean square error (MSE) between the generated sign pose sequence and the target sequence:

$$\mathcal{L}_{\text{gen}} = \text{MSE}(\mathbf{Y}_{\text{gen}}, \mathbf{Y}) \tag{2}$$

3.2 External Aligner

We next introduce the External Aligner, a separate model from our NAT model, which generates target durations for the duration predictor. In order to get the target duration, the EA requires to learn the alignment between the gloss sequence and the corresponding sign pose sequence. Due to the error accumulation problem, it is difficult to extract satisfactory alignment from a well-trained autoregressive model. Hence, we first apply a mix density network [5] to transform the input gloss sequence $\mathbf{X}$ into the gloss-synchronized sign pose Gaussian distribution $\mathbf{Z} = \{\mathbf{z}_i\}_{i=1}^{N_X}$, where $\mathbf{z}_i = (\mu_i, \sigma_i)$ is the mean and variance of the Gaussian distribution of the sign pose frame that is correspond to the i-th gloss. Concretely, the mix density network contains a FFT block, followed by two linear layers that project the hidden states into the mean sequence and the variance sequence of the gloss-synchronized sign pose Gaussian distribution sequence respectively. The FFT block is not shareable with that in the NAT model, which makes the EA more flexible.

We define the correspondence between the sign pose frame and the sign pose Gaussian distribution as an alignment A. Thus, if the target sign pose frame y_j follows the predicted distribution of i-th gloss $\mathcal{N}(y_j; \mu_i, \sigma_i)$, then we define $A(j) = i$. Next, we apply the monotonic alignment search [25] to get the most probable monotonic alignment A^* between the gloss-synchronized sign pose Gaussian distribution sequence Z and the target sign pose sequence Y, which maximizes the log-likelihood of the data:

$$\begin{aligned} A^* &= \arg\max_A \log P_{(\mathcal{X};\mathcal{Y})}(Y \mid X; A, \theta) \\ &= \arg\max_A \sum_{j=1}^{N_Y} \log \mathcal{N}\left(y_j; \mu_{A(j)}, \sigma_{A(j)}\right), \end{aligned} \tag{3}$$

where $(\mathcal{X}; \mathcal{Y})$ is the dataset and θ is the parameters of the mix density network. Then we can optimize the mix density network via the alignment loss defined as:

$$\begin{aligned} \mathcal{L}_{align} &= -\log P_{(\mathcal{X};\mathcal{Y})}(Y \mid X; A^*, \theta) \\ &= -\sum_{j=1}^{N_Y} \log \mathcal{N}\left(y_j; \mu_{A^*(j)}, \sigma_{A^*(j)}\right) \end{aligned} \tag{4}$$

After training, the target logarithmic durations of the gloss sequence can be calculated from the most probable monotonic alignment A^* as follows:

$$l_i^{trg} = \log \sum_{j=1}^{N_Y} 1_{A^*(j)=i}, i = 1, \cdots, N_X, \tag{5}$$

where l_i^{trg} is the target logarithmic interval of i-th gloss.

3.3 Spatial-Temporal Graph Convolutional Pose Generator

Table 1: Design of the STGCPG, where LR is shorthand for LeakyReLU, operator Conv_{st} is the spatial-temporal graph convolution, operator Up_{s} is the graph upsampling operator.

Block	Operations	Output Dimensions
In	Pose-synchronized feature sequence	(T, 1, 256)
0	LR ∘ BN ∘ Conv1d	(T, 1, 512) viewed as (T, 2, 256)
1	LR ∘ BN ∘ Conv_{st} ∘ Up_{s}	(T, 13, 256)
2	LR ∘ BN ∘ Conv_{st} ∘ Up_{s}	(T, 25, 64)
3	LR ∘ BN ∘ Conv_{st} ∘ Up_{s}	(T, 50, 16)
Out	Conv_{st}	(T, 50, 3)

In this section, we introduce the Spatial-Temporal Graph Convolutional Pose Generator (STGCPG), which transforms pose synchronized feature sequences into sign pose sequences. The architecture of the STGCPG is outlined in Table 1, where the sizes of spatial-temporal graphs are denoted as (T, V, C). To generate smoother and more natural sign pose sequences, we introduce two graph operations: graph upsampling and spatial-temporal graph convolution, which will be illustrated in the following.

Given a pose-synchronized feature sequence $H_{\text{pose}} \in \mathbb{R}^{N_Y \times 1 \times C}$, where C is the feature dimension and it has no spatial extent. We aim to transform it into the corresponding sign pose sequence $Y_{gen} \in \mathbb{R}^{N_Y \times J \times K}$, which can be represented as a spatial-temporal

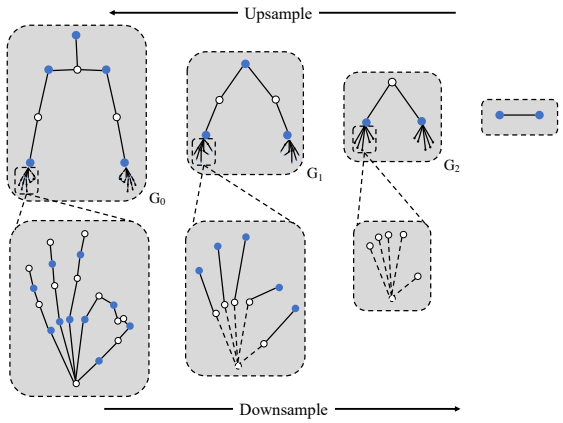


Figure 3: Graph pyramid for the PHOENIX14T dataset. G_0 is the original sign pose graph. Graph pyramids provide a unambiguous guidance for performing graph upsampling.

graph, which is a temporal sequence of sign pose graphs, and one sign pose graph is shown as G_0 in Figure 3. One proven idea in image generation is to perform gradual upsampling [34], which increases the resolution and adds details in a layer-by-layer manner. However, as sign pose graphs are not regular grids, it is nontrivial to devise a path which we can follow to perform gradual upsampling. Inspired by Yan and Li [49], we develop a sign pose graph pyramid to solve this problem. The graph pyramid is constructed from a full resolution sign pose graph with J joints. An example graph pyramid built on the sign pose graph of PHOENIX14T dataset [7] is shown in Figure 3. As the example shows, we perform graph subsampling by removing all joints colored white, which are separated by blue joints that are reserved. We repeat this operation until there are only two joints left. The result is a graph pyramid, where the i-th level of graph G_i is the output of the i-th iteration of subsampling. Graph upsampling $\mathrm{Up_s}$ is the inverse operation of subsampling, which can be implemented by retracing the path from the top of the pyramid, the two joints graph, to the bottom, the full graph. Concretely, we first embed the coarser graph into the finer graph according to the pyramid and then assign values to any new joint by averaging the values of its neighboring joints.

Spatial-temporal graph convolution is widely-used in skeleton-based action generation and recognition [49, 50], which usually combines spatial graph convolution and temporal convolution to model spatial-temporal information effectively. In this work, we adopt this architecture in sign pose sequence generation. Following the practice of Yan and Li [49], the parameters of $\mathrm{Conv_{st}}$ networks in locally connected blocks are not shared across joints, which introduces extra degree of freedom to our STGCPG.

3.4 Training of NAT-EA

Based on the aforementioned model design, we first train the EA by optimizing the alignment loss defined in Equation 4. Then we can use the well-trained EA to generate the target logarithmic durations of the gloss sequence by Equation 5, and train the NAT model in an end-to-end manner with a multi-task loss, given by:

$$\mathcal{L}_{\mathrm{NAT}} = \mathcal{L}_{\mathrm{gen}} + \lambda \mathcal{L}_{\mathrm{dur}}, \tag{6}$$

where λ is the hyper-parameter to control the balance of losses.

4 EXPERIMENTS

4.1 Dataset

To the best of our knowledge, there is only one public SLP dataset that is suitable for the typical SLP pipeline—the RWTH-PHOENIX-Weather2014T [7], a German sign language dataset, which contains 8257 weather forecasts samples from 9 signers. It has two-stage annotations: sign gloss annotations with a vocabulary of 1066 different signs for Glosses to Poses generation and German translation annotation with a vocabulary of 2877 different words for Text to Glosses translation. Additionally, it has been officially split into 7096, 519 and 642 samples for training, validation, and testing respectively. We conduct extensive experiments on this dataset.

4.2 Evaluation Criteria

Following the widely-used setting in SLP [40], we apply back-translation as a means of evaluation. We use the state-of-the-art SLT [8] as our back translation model, modified to take sign pose sequences as input. It is trained on the PHOENIX14T dataset. We generate spoken language translations and glosses recognition results of the produced sign pose sequences and compute BLEU, ROUGE and chrF [33] scores and Word Error Rate (WER) [40] respectively. We provide BLEU n-grams from 1 to 4 for completeness.

Additionally, we propose to report results according to the average Euclidean distance between predicted joints and ground truth in Euclidean space (Mean Joint Error), a widely accepted metric for human motion prediction [28]. Since the generated sign pose sequence is usually different from the ground truth in length, we apply Dynamic Time Warping (DTW) [4] to align them before computing Mean Joint Error, and we call this criterion DTW-MJE.

4.3 Implementation Details

We next introduce the implementation details of our NAT-EA model and the back-translation model.

Data Preprocessing. Since PHOENIX14T dataset itself doesn't contain any pose information, we need to generate the ground truth sign pose sequences. Following B. Saunders et al. [40], we first extract 2D joint positions from each video using OpenPose [9], Then we apply the skeletal model estimation improvement method presented in [54] to lift the 2D joint positions to 3D whilst maintaining consistent bone length and correcting misplaced joints. At last, we apply skeleton normalisation similar to [42] to remove the skeleton size difference between different signers and represent 3D joints as x, y and z coordinates.

Model Setting. The FFT blocks in NAT model are built with 2 layers, 8 heads and embedding size of 256, which is the same as that in previous work [40] for a fair comparison. The FFT block in the mix density network is built with 4 layers, 8 heads and embedding size of 256. The temporal and spatial kernel size of $\mathrm{Conv_{st}}$ are set to 9 and 3 respectively. During training of the NAT model, we set λ

Table 2: Quantitative results for G2P task on PHOENIX14T test set. '†' indicates the model is tested by us under a fair setting.

Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	chrF	ROUGE	WER	DTW-MJE
Mean	3.06	0.94	0.33	0.11	11.61	5.05	99.39	0.154
Random	4.48	1.51	0.51	0.22	12.44	5.58	98.99	0.214
Autoregressive Models								
PT(base)†	6.27	3.33	2.14	1.59	14.52	9.50	96.86	0.197
PT(FP&GN)†	11.45	7.08	5.08	4.04	19.09	14.87	94.65	0.191
Non-Autoregressive Models								
NAT-AT	14.26	9.93	7.11	5.53	21.87	18.72	88.15	0.177
NAT-EA w/o. STGCPG	**15.13**	10.31	7.90	6.61	22.70	**19.96**	85.40	0.159
NAT-EA	15.12	**10.45**	**7.99**	**6.66**	**22.98**	19.43	**82.01**	**0.146**
GT	24.12	16.77	12.80	10.58	30.63	27.70	55.93	0.0

to 1.0 for simplicity. And we use an Adam optimizer [14] with the initial learning rate 0.001 and batch size 32.

4.4 Quantitative Evaluation

We compare our NAT-EA method with several other models, including the autoregressive state-of-the-art method, and other baseline models.

4.4.1 Baselines.

Always predict the mean pose. Assuming that the motion extents of signers are not large, we use the mean pose of the training set as generated sign pose, and use the lengths of the target sign pose sequences as generated sequence lengths.

Predict a randomly chosen pose. In this baseline, we randomly select a different pose sequence from the training set and use this as our prediction. The generated sequence lengths are from the target sign pose sequences. Since these are real signer poses, there is reason to think it would generate qualitatively appealing motion [18].

Progressive Transformer [40]. It is the first SLP model to tackle the G2P problem of the typical SLP pipeline in an autoregressive manner. Since B. Saunders et al. [40] didn't release their pre-trained back-translation model, and they use the ground truth first sign pose frame and timing information at inference, their presented results are not comparable to ours. We reproduce the results of its base model, **PT(base)**, and its model with future prediction and Gaussian noise, **PT(FP&GN)**, without extra ground truth information at inference, and by our back-translation model.

4.4.2 Main Results. The overall evaluation results on PHOENIX14T dataset are presented in Table 2, where the GT represents the evaluation results of our back-translation model. The results reveal some interesting points:

- Our NAT-EA method achieves the best performance on almost all criteria, especially on WER and DTW-MJE score, and outperforms autoregressive methods, PT(base) and PT(FP&GN), with a notable margin. This fact verifies the effectiveness of our NAT-EA method.
- The DTW-MJE scores of autoregressive methods are far large than that of mean pose, which verifies the negative effects of the error accumulation problem to some extent.
- On WER and DTW-MJE score, PT(FP&GN) only exceeds PT(base) with a slight margin. This fact verifies that it cannot resolve the error accumulation problem effectively only by data augmentation approaches without a non-autoregressive framework.

4.5 Ablation Study

In this section, we conduct the ablation study to verify the effectiveness of our model design, including the External Aligner (EA) and the Spatial-Temporal Graph Convolutional Pose Generator (STGCPG).

The Effectiveness of EA. We remove the EA and utilize a well-trained autoregressive model, PT(FP&GN), to get the alignments from the decoder-to-encoder attention as FastSpeech [37] does. This method is termed as NAT model with AuToregressive model (**NAT-AT**).

The Effectiveness of STGCPG. We remove the STGCPG in the NAT model, and apply a linear projection layer to generate sign pose sequences. This method is termed as **w/o. STGCPG**.

The ablation results on PHOENIX14T dataset are reported in Table 2 and we can find some interesting points:

- The full model achieves better results than model (w/o. STGCPG). It suggests that the STGCPG module can make the generated sign pose sequence closer to the ground truth and improves back-translation performance.
- The NAT-AT model has severe performance degradation than the NAT-EA model. This fact demonstrates that the proposed External Aligner can provide better alignments than the autoregressive model, which suffers severe error accumulation problem.

4.6 Analysis

4.6.1 Error Accumulation Verification. Li et al. [28] has proven the existence of the error accumulation problem in autoregressive models for human motion prediction. Here we conduct a experiment to further verify this fact in SLP. Specifically, we analyse all sign pose sequences generated by a well-trained autoregressive model, PT(base), and a well-trained non-autoregressive model, NAT-EA, on PHOENIX14T test set. We first apply DTW to align the generated sign pose sequence and the ground truth. Then we record the

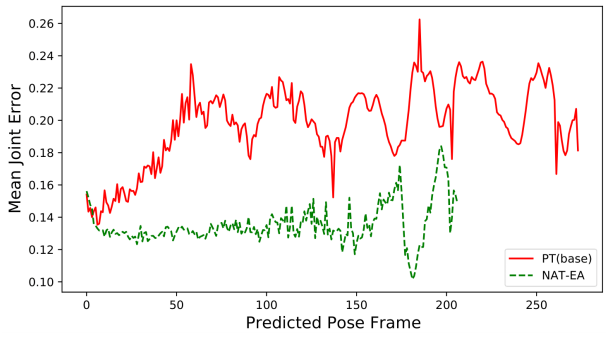


Figure 4: Error accumulation verification results of PT(base) and NAT-EA.

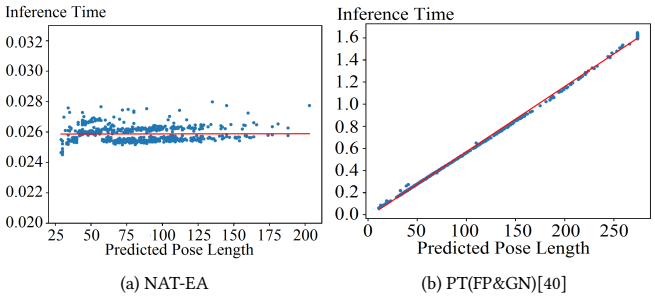


Figure 5: Relationship between inference time (second) and predicted sign pose sequence length for NAT-EA and PT(FP&GN).

average single frame MJE between the aligned sign pose sequence and the ground truth at different frames of the original generated sign pose sequence. The results are shown in Figure 4. We note that the MJE of the autoregressive model increases approximately linearly while the MJE of our proposed NAT-EA does not increase when the frame number less than 50. During the later half, the MJEs of both models begin to fluctuate due to data deficiency. This fact verifies that autoregressive models for SLP also suffer the error accumulation problem, and our proposed NAT-EA model resolve this problem effectively.

4.6.2 Speedup Analysis. We compare the average inference latency of NAT-EA with that of the autoregressive SLP model. And then, we analyze the relationship between speedup and the length of the predicted sequence.

Average Latency Comparison. The average latency is measured in average time in seconds required to generate one sign pose sequence on the test set of PHOENIX14T dataset. We record the inference latency and corresponding back-translation WER of PT(FP&GN) [40] and NAT-EA, which is listed in Table 3. The result shows that NAT-EA speeds up the inference by 21.8× on average, compared with PT(FP&GN) which has similar number of model parameters.

Table 3: The comparison of average inference latency and corresponding back-translation WER between the PT(FP&GN) and our proposed NAT-EA.

Method	WER	Latency (s)	Speedup
PT(FP&GN)[40]	94.65%	0.566	1×
NAT-EA	82.01%	0.026	21.8×

Relationship between Speedup and Length. We visualize the relationship between the inference latency and the length of the generated sign pose sequence in Figure 5. It can be seen that the inference latency increases approximately linearly with the generated sign pose sequence length for PT(FP&GN), while nearly holds a small constant for NAT-EA. Then, we bucket the test sign pose sequences of length within [175, 200], and calculate their average inference latency for PT(FP&GN) and NAT-EA to obtain the maximum speedup on PHOENIX14T test set. The results are 1.071s and 0.026s for PT(FP&GN) and NAT-EA respectively, which shows that NAT-EA achieves the speedup up to 40.97× on PHOENIX14T test set, thanks to the parallel generation which is insensitive to sequence length.

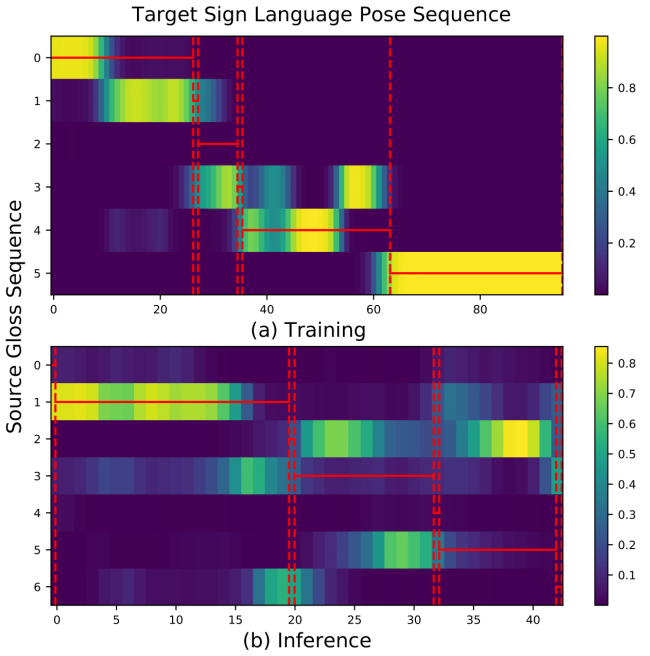


Figure 6: An example of the visualization of predicted durations during training and inference.

4.6.3 The Visualization of Predicted Durations. We visualize the encoder-decoder attention map in Figure 6, which is obtained from the well-trained autoregressive model, PT(FP&GN). The attention maps illustrate the alignments between source gloss sequences and the corresponding target sign pose sequences during training and inference. The horizontal red lines are the predicted durations of the well-trained EA during training and inference.

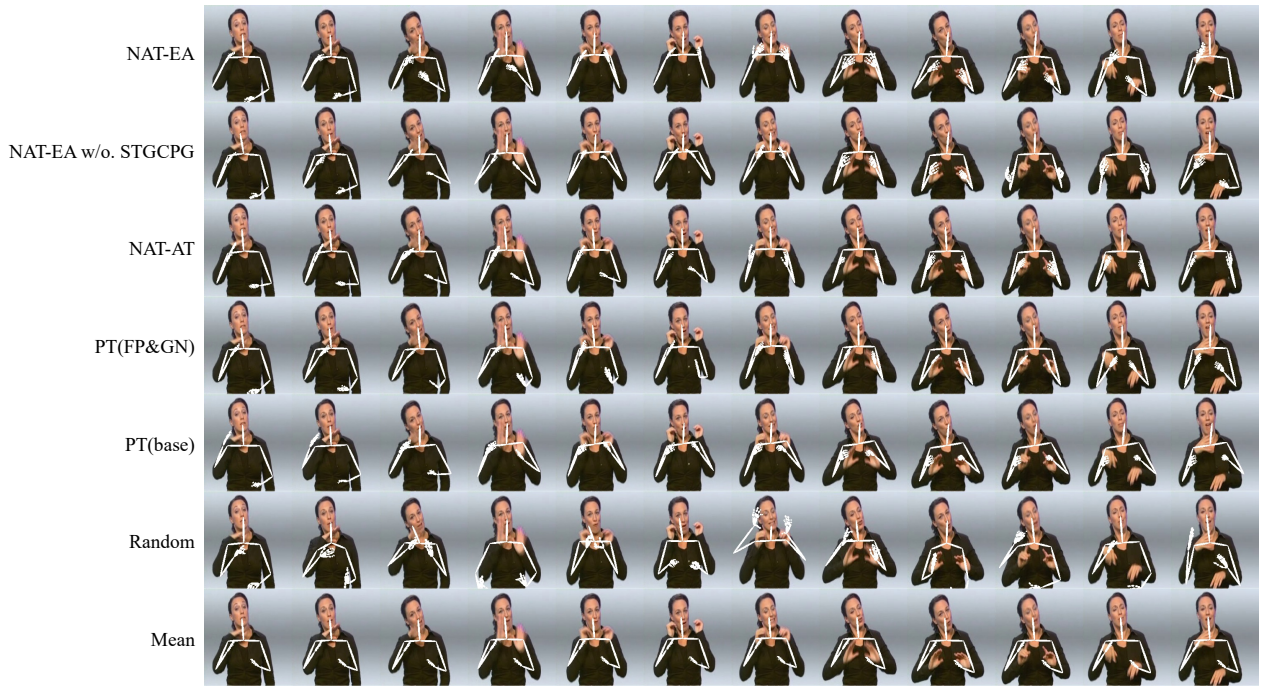
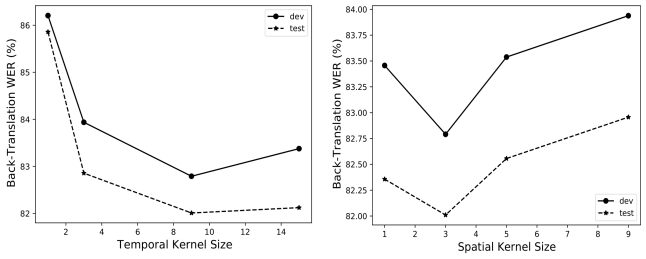


Figure 7: G2P qualitative results. We show the generated sign poses overlaid on the ground-truth video. See our supplementary material for more results.

The figure 6(a) shows that the horizontal red lines roughly correspond to the attention map during training when the autoregressive model can get satisfactory alignments due to teacher forcing. It means that the EA makes a right prediction of gloss durations.

The figure 6(b) shows that, the horizontal red lines conform to the attention map only at the beginning of the sign pose sequence during inference. It verifies the existence of the error accumulation problem in the autoregressive model.

4.6.4 Hyper-Parameters Analysis. In NAT-EA, the spatial and temporal kernel size of STGCPG, k_s and k_t, are important hyper-parameters. Therefore, we further explore their effects. We first set $k_s = 3$ to see the effect of k_t, and then we set $k_t = 9$ to see the effect of k_s, as shown in Figure 8. It can be seen that, when setting $k_s = 3, k_t = 9$, the back-translation accuracy peaks.



(a) Temporal kernel size (b) Spatial kernel size

Figure 8: Effect of different temporal and spatial kernel size in STGCPG.

4.7 Qualitative Results

We qualitatively compare our NAT-EA method to the baselines and the ground truth sign pose sequence on PHOENIX14T test set in Figure 7. Please refer to our supplementary video results which better convey temporal information.

5 CONCLUSION

In this work, we verify the existence of the high inference latency and the error accumulation problem in previous autoregressive models for G2P task of the typical SLP pipeline, and propose a novel Non-AuToregressive model with External Aligner (NAT-EA) to circumvent these problems. We extract alignments from an external aligner by monotonic alignment search for gloss duration prediction, which is used by a length regulator to expand the source gloss sequence to match the length of the target sign pose sequence for parallel sign pose generation. Furthermore, we devise a spatial-temporal graph convolutional pose generator in the NAT model to generate smoother and more natural sign pose sequence. Extensive experiments conducted on PHOENIX14T dataset show that our proposed model significantly outperforms previous autoregressive models in terms of speed and quality.

ACKNOWLEDGMENTS

This work was supported in part by the National Key R&D Program of China under Grant No.2020YFC0832505, National Natural Science Foundation of China under Grant No.61836002, No.62072397, No.62077041 and Zhejiang Natural Science Foundation under Grant LR19F020006.

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